

Probability Lecture 1

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Sample Space

A set of all possible outcomes(Ω)

Set must be

- Mutually exclusive
- Collectively exhaustive
- At the "right" granularity

Probability Axioms

Event: A subset of the sample space

Probabilities are assigned to events

The Axioms are

1. Nonnegativity $P(A) \geq 0$
2. Normalization $P(\Omega) = 1$
3. (Finite) Additivity, If $A \cap B = \emptyset$ then $P(A \cup B) = P(A) + P(B)$

Simple Consequences of the Axioms

We know,

$$A \cup A^c = \Omega$$

and,

$$A \cap A^c = \emptyset$$

Thus from our axioms we get,

$$1 = P(\Omega) = P(A \cup A^c) = P(A) + P(A^c)$$

Also,

$$P(A) = 1 - P(A^c) \leq 1$$

Finally,

$$1 = P(\Omega) + P(\Omega^c)$$

$$1 = 1 + P(\emptyset)$$

$$P(\emptyset) = 0$$

Strengthening Axiom 3

$$P(A \cup B \cup C) = P((A \cup B) \cup C) = P(A \cup B) + P(C) = P(A) + P(B) + P(C)$$

Thus we can see that,

If $A_1, \dots, A_k$ are disjoint events $\Rightarrow P(A_1 \cup \dots \cup A_k) = \sum_{i=1}^k P(A_i)$

Further Consequences

Let $A \subset B$,

$$B = A \cup (B \cap A^c)$$
$$P(B) = P(A) + P(B \cap A^c) \geq P(A)$$

Let A, B be two sets such that $P(A \cap B) \neq \emptyset$

We want $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Let,

$$a = P(A \cap B^c), \quad b = P(A \cap B), \quad c = P(B \cap A^c)$$

Then,

$$P(A \cup B) = a + b + c$$
$$P(A) + P(B) - P(A \cap B) = (a + b) + (b + c) - b = a + b + c$$

Since $P(A \cap B) \geq 0$, $P(A \cup B) \leq P(A) + P(B)$ (union bound)

Finally,

$$A \cup B \cup C = A \cup (B \cap A^c) \cup (C \cap A^c \cap B^c)$$

These events are disjoint, thus

$$P(A \cup B \cup C) = P(A) + P(B \cap A^c) + P(C \cap A^c \cap B^c)$$

Discrete Uniform Law

Assume Ω contains n equally likely elements, assume the event A consists of k elements.

Then

$$P(A) = k \cdot \frac{1}{n}$$

If Ω is not discrete, Probability = Area

Probability Calculation Steps

- Specify the sample space
- Specify a probability law
- Identify an event of interest
- Calculate...

Countable Additivity Axiom

If $A_1, A_2, A_3 \dots$ is an infinite sequence of disjoint events then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

Note

This only holds for "countable" sequences of events (shouldn't be a problem in this module)

Using this Axiom on the unit square we see that it doesn't hold

$$1 = P(\Omega) = P(\cup\{(x, y)\}) = \sum P(\{(x, y)\}) = \sum 0 = 0$$